De Broglie's Matter Waves

- Compton's equation demonstrated that EM waves could behave like particles.

Following the reasoning of Planck & Einstein

De Broglie's Speculative Hypothesis ⇒ electrons & other particles of matter behave like waves

De Broglie's Matter Waves

De Broglie's Hypothesis

⊕

Bohr's Quantum Theory

} WAVE QUANTUM MECHANICS

⇒ According to De Broglie's hypothesis, massless photons as well as massive particles must satisfy a common set of relations that connect energy, frequency, linear momentum and wavelength.

E, f, p, λ

De Broglie's Matter Waves

- Any particle that has energy and momentum is a de Broglie wave that can be described as follows:

$$E = hf \quad \& \quad \lambda = \frac{h}{p}$$

↑ relativistic → p

energy & momentum

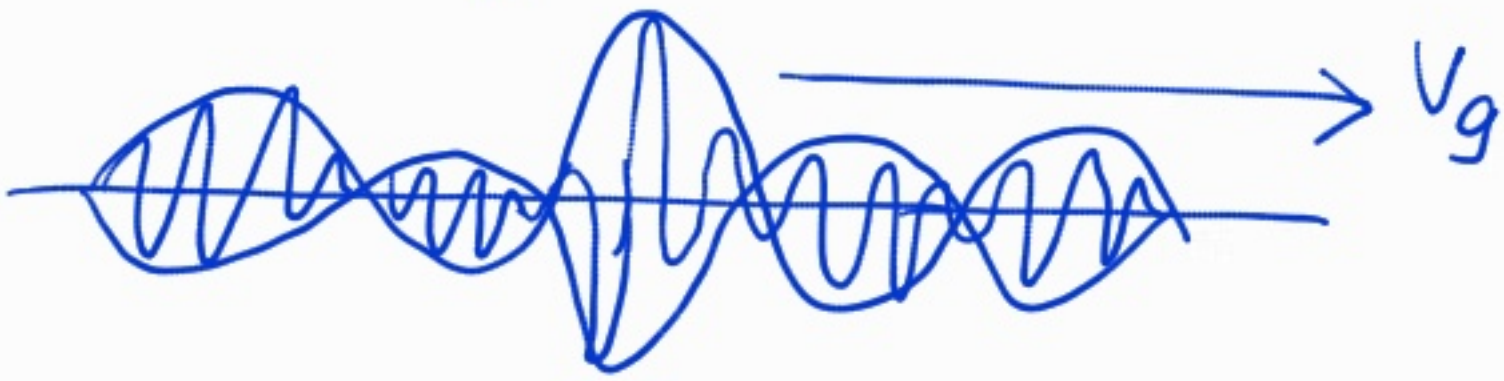
- Since $k = 2\pi/\lambda$ & $\omega = 2\pi f$

$$E = \hbar\omega \qquad \vec{p} = \hbar\vec{k} \qquad \hbar = \frac{h}{2\pi}$$

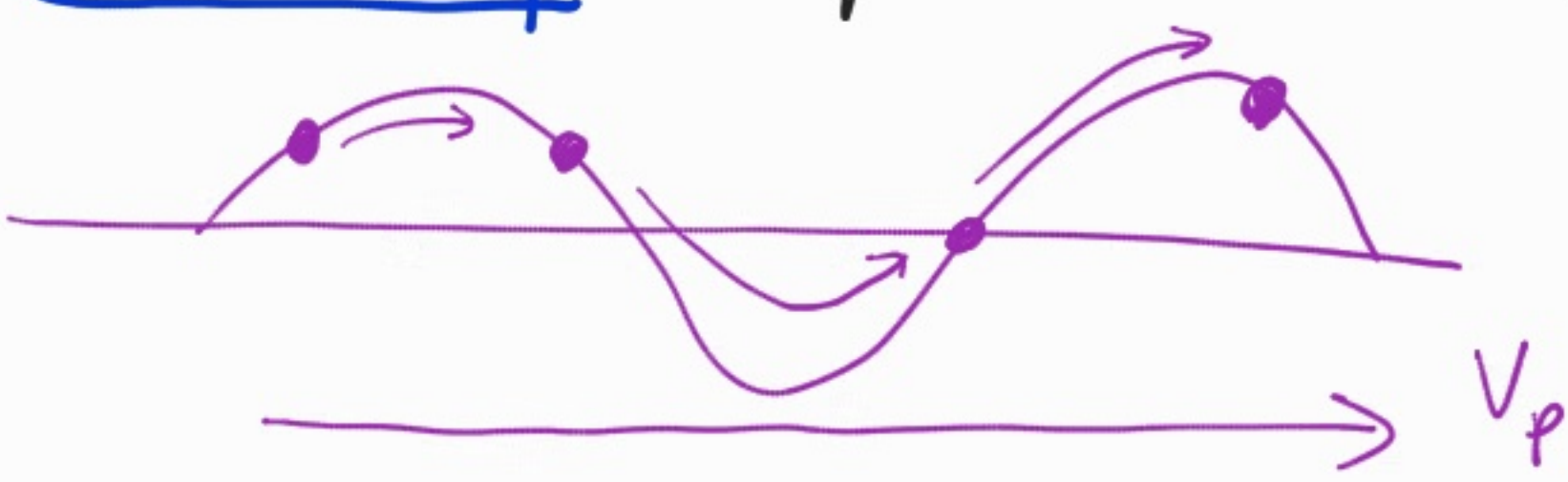
De Broglie's Matter Waves

- de Broglie's wave travels with group velocity.

Group Velocity → velocity of overall envelope shape of the waves amplitude



Phase Velocity → velocity at which the phase of any one f component travels.



De Broglie's Matter Waves

- Group velocity = velocity of the particle
  u (m/s)

- Considering relativistic energy $E = mc^2$ and relativistic momentum $p = mu$, the following De Broglie's relation are obtained:

$$\lambda f = \frac{\omega}{k} = \frac{E/\hbar}{p/\hbar} = \frac{E}{p} = \frac{mc^2}{mu} = \frac{c^2}{u}$$

Define $\frac{c}{\beta} = \frac{c^2}{u}$, $\therefore \beta = \frac{u}{c}$ [velocity ratio]

- If particle is massless, then $u = c$:

$\therefore \lambda f = c$ is recovered

De Broglie's Matter Waves

- Let us determine the de Broglie wavelength of a non-relativistic electron with :

- Recall : according to the Theory of Special Relativity, the total energy of any particle in motion is given by :

$$E^2 = p^2c^2 + m_0^2c^4$$

(m_0 ← rest energy)

relativistic energy equation for photon momentum

De Broglie's Matter Waves

$$p^2c^2 = E^2 - m_0^2c^4 = E^2 - E_0^2$$

Kinetic Energy $\Rightarrow K$

$$K = E - E_0 \longrightarrow E = K + E_0$$

$$p = \sqrt{\frac{E^2 - E_0^2}{c^2}}$$

$$= \sqrt{\frac{K(K + 2E_0)}{c}}$$

$$\lambda f = \frac{E}{p} \rightarrow \lambda = \frac{E/f}{p} = \frac{h}{p}$$

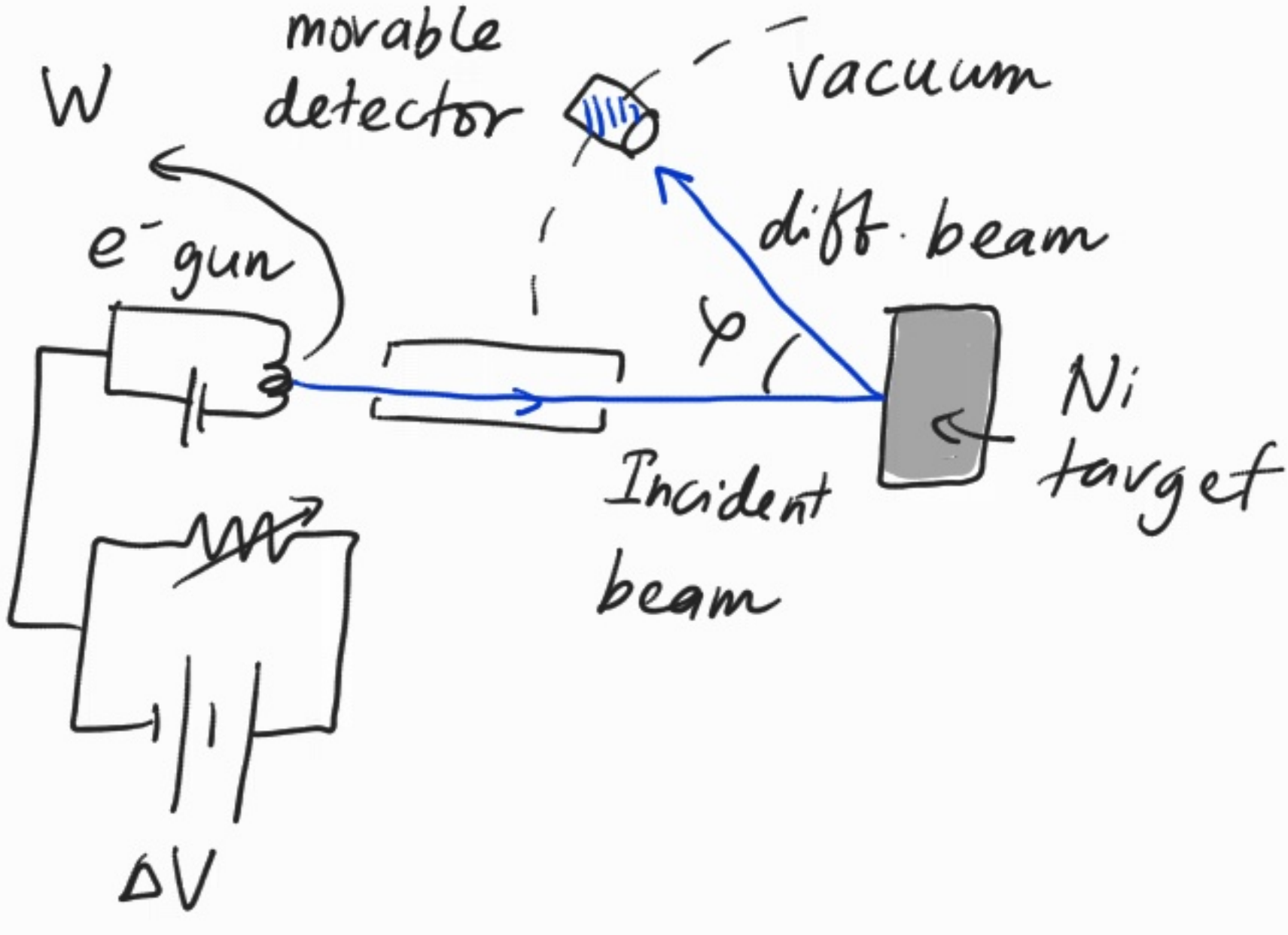
$$\therefore \lambda = \frac{h}{p} = \frac{hc}{\sqrt{K(K + 2E_0)}}$$

Electron Diffraction

- Davisson & Germer performed a series of electron scattering experiments.
- The experiments conclusively showed that electrons display wave-like behavior.

↳ although they did not design their experiment to prove/disprove de Broglie's hypothesis

Electron Diffraction



Davisson - Germer Diffraction Experiment

Electron Diffraction

- Ni sample was prepared in a high T oven to regularize its polycrystalline structure.
- Thermal electrons are released from the heated element (made of tungsten, W) from an electron gun.
- Kinetic energy of e^- are controlled by the potential difference, ΔV.

$$e\Delta V = K = \frac{p^2}{2m} \quad \text{where}$$

$$p = \sqrt{2me\Delta V}$$

Electron Diffraction

① Polycrystalline Structure → random crystal orientation

② Regular crystalline Structure → regular crystal orientation

Case ①

⇒ e⁻ are scattered on random directions

⇒ looks like light reflecting off a porous surface.

Electron Diffraction

Case (2)

⇒ intensity of scattered e^- show a clear maximum at a specific angle.

⇒ results show a clear diffraction pattern

Note:

- A similar result was observed by W.L. Bragg and W.H. Bragg in X-ray crystallography.

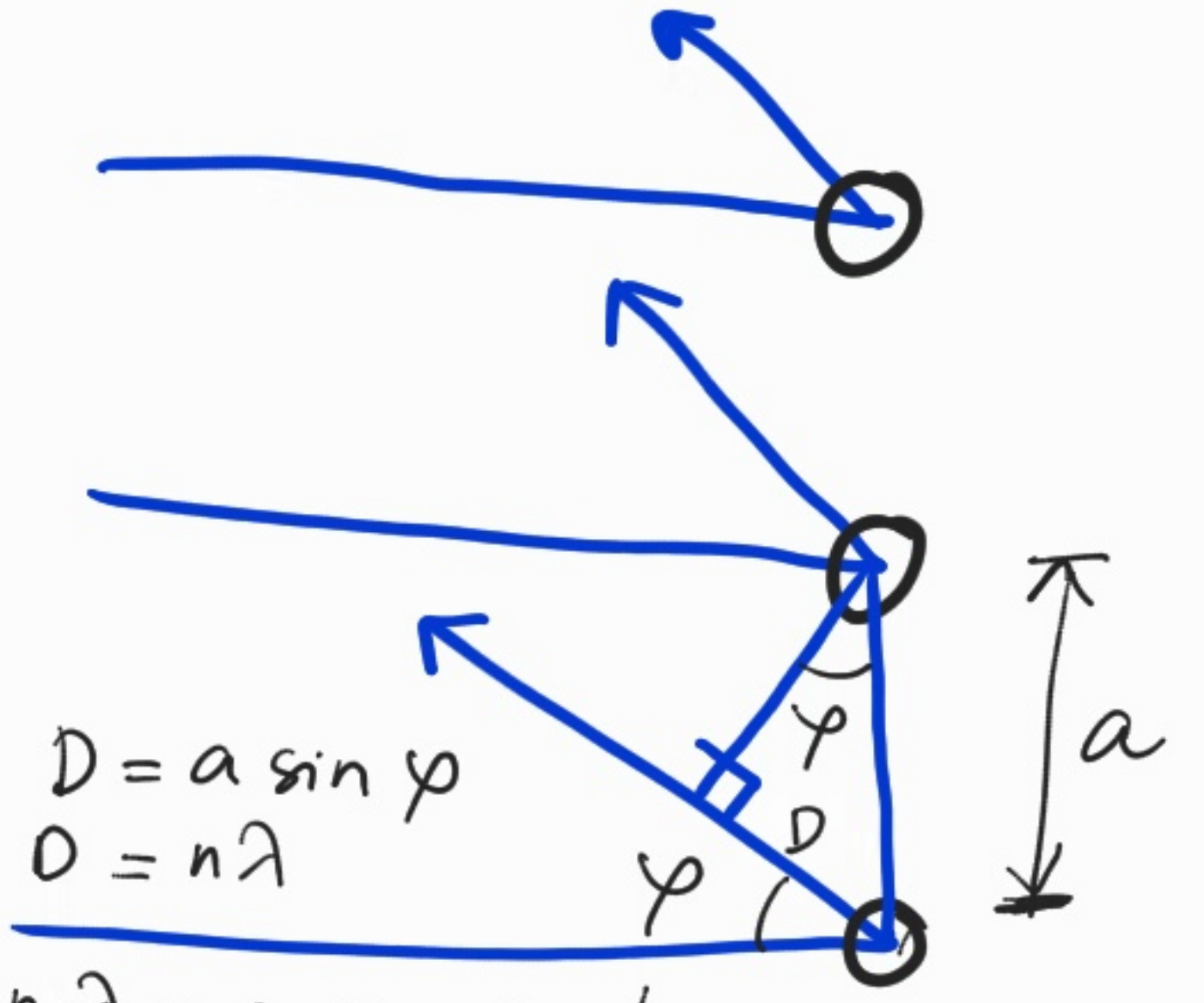
λ Connection ⟷ crystal lattice spacing

Electron Diffraction

Crystalline Structure of Ni

↳ X-ray crystallography to determine lattice spacing, a

$a = 2.15 \text{ Å}$



$n\lambda = a \sin\varphi$ for $n = 1, 2, 3, \ldots$

Practice Problem 1

Consider the energy states of an electron receiving a photon with a frequency 2×10^{12} Hz. The eletron then becomes excited and changes its energy level. Determine the difference in the electron's energy levels once it receives the photon in J.

Planck's constant:

$$h = 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$$

Solution.

1/ Energy states = difference between the quantum states

2/ Quantum number for energy levels, $n \longrightarrow n+1$

$\therefore \Delta E = E_{n+1} - E_n$

$= (n+1)hf - nhf$

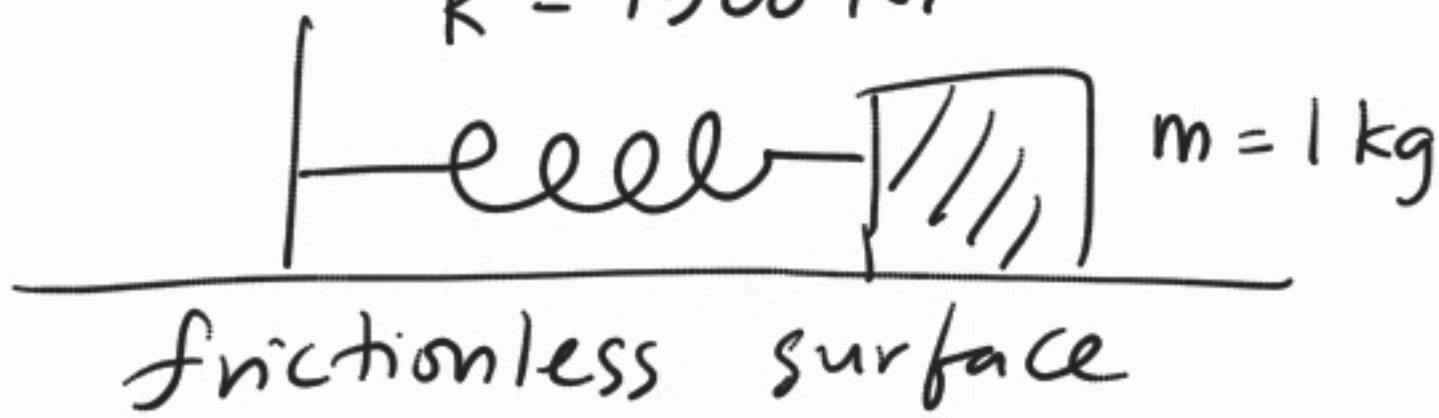
$= hf$

$= (6.626 \times 10^{-34}\ J.s) \times (2 \times 10^{12}\ s^{-1})$

$= 13.252 \times 10^{-22}\ J$

Practice Problem 2

Consider the quantization of a classical oscillator: a spring-mass system.



If the amplitude of these oscillations are 0.5m, determine the difference between the energy levels of the oscillator in J } Compare it with overall macroscopic energy of the system.

Strategy.

1/ Determine f using Hooke's Law:

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

2/ Calculate the energy quantum.

$$\Delta E = hf$$

3, Determine Spring potential energy (macroscopic energy)

$$E' = \frac{1}{2} k A^2$$

Solution.

$$f = \frac{1}{2\pi}\sqrt{\frac{k}{m}} = \frac{1}{2\pi}\sqrt{\frac{1500\ N/m}{1\ kg}}$$

$$= 6.164\ Hz$$

$$\Delta E = (6.626\times10^{-34}\ J.s)(6.164\ Hz)$$

$$= 40.84\times10^{-34}\ J$$

Macroscopic Energy:

$$E' = \frac{1}{2}(1500\ N/m)(0.5m)^2$$

$$= 187.5\ J$$

$$\frac{\Delta E}{E'} \approx 10^{-35}$$

} quantum effects are insignificant.